Topological properties of electronic bands in 2D materials

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In the last thirty years, topology has become increasingly prevalent in condensed matter physics. The new paradigm has enabled theorists to develop precise and elegant descriptions of a broad range of exotic systems. More recently, an emerging field known as twistronics, which involves the twisting of 2D material layers to create moiré patterns, has received considerable attention. The ability to host correlated insulating states and unconventional superconductivity makes these systems an excellent platform for further research. This brief review provides an introduction to topology and moiré theory, and explains how they apply to condensed matter systems. To further demonstrate the importance of topological properties of electronic bands in 2D materials, two topologically interesting physical systems are introduced: twisted double bilayer graphene (tDBG) and twisted bilayer graphene on nearly aligned hexagonal boron nitride (tBG/hBN).

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I. INTRODUCTION

The rich - and often surprising - variety of emergent phenomena in condensed matter systems presents a considerable challenge for those seeking consistent and predictive theoretical descriptions. On the basis of spontaneous symmetry breaking, Landau provided a general framework under which different phases of matter can be understood, including charge-density waves, ferromagnetism, antiferromagnetism and superconductivity. The evolution of the *local* order parameter in Landau's theory can be used to predict and classify properties of a given phase transition. In the last thirty years, beginning with the quantum Hall effect, systems harbouring *topological* order have been discovered, where *global* parameters dictate the observed phases. This can lead to robust states, unaffected by lattice imperfections and disorder. Today, topology in condensed matter systems is ever more common - spin liquids [1], topological insulators [2] and potential applications in fault-tolerant quantum computing [3] are a few important examples.

In the context of band theory, which was conceived almost a century ago from Bloch's theorem, topological band theory remains a new and evolving paradigm: its influence and importance is likely yet to be fully understood. This is particularly true for its role in 2D materials, an even more immature field.

The successful isolation of single layer graphene through the exfoliation of graphite [4] was arguably the inception of the 2D materials field. Since then, a large and ever-expanding family of 2D crystals have been isolated and tested. Through stacking of atomic layers, van der Waals (vdW) heterostructures [5] can be formed, allowing electronic properties to be tuned. An additional degree of freedom, the relative orientation between 2D layers, allows for further control. A *Moiré* pattern is formed, which can result in highly correlated states at a magic twist angle. Recently, correlated insulating and unconventional superconducting states were observed in twisted bilayer graphene (tBG) [6, 7]. As a consequence, there has been a concentrated effort to explore related graphene Moiré systems, with the prospect of furthering fundamental understanding, but also of discovering new, exotic phases of matter. Two of these related systems are explored in this work.

This article is structured as follows: in Section II, a background on topology and its connections with the Berry curvature is presented, as well as an overview of graphene and Moiré physics in tBG. Section III focuses

on twisted double bilayer graphene, how it may be modelled in a continuum theory and its corresponding electronic and topological properties. Section IV explores a twisted bilayer graphene (tBG) on hexagonal boron nitride (hBN) superlattice system, and discusses how the presence or the absence of a Quantum Anomalous Hall Effect (QAHE) can be related to the commensurability of the structure. Finally, Section V provides a future outlook and a brief conclusion.

II. BACKGROUND

A. Introduction to Topology

Topology is the abstract study of continuity and it focuses on the features of structures that remain the same when continuous changes are made to the parameters of the system. It has become increasingly prevalent in condensed matter these days, and despite its apparent unexpected connection to physics, it has provided the framework to understand and predict a variety of novel physical phenomena [8].

In geometric topology, continuous deformations refer to stretching, twisting or bending, but without closing or opening holes in the geometric object. Since under these transformations the number of holes, also called genus, remains unchained, we refer to it as a topological invariant.

Similarly, when discussing band structure topology, continuous deformations refer to all the smooth transformations of the Hamiltonian which do not open nor close any gaps in the energy dispersion corresponding to it. Now our topological invariant is called a Chern number of the first class, $C_1 \in \mathbb{Z}$, and therefore going from one number to another would correspond to a phase transition or a quantum critical point [8].

Chern numbers made their first appearance in condensed matter physics in the famous Thouless-Kohmoto-Nightingale-den Nijs (TKNN) paper on the quantum Hall effect [9], and later led to major breakthroughs in the field, which were awarded the 2016 Nobel prize.

B. Berry Phase, Connection and Curvature

The concept of Berry phase is central in all discussions of topological aspects in the field of condensed matter physics. We start by considering a Hamiltonian which depends on time through a set of parameters $\boldsymbol{\mathcal{R}}(t) = \{\mathcal{R}_1(t), \mathcal{R}_2(t), ...\mathcal{R}_N(t)\}$, and we are interested in its (adiabatic) evolution as $\boldsymbol{\mathcal{R}}(t)$ moves slowly along a path $\mathcal{C}$ in parameter space.

For an instantaneous orthonormal basis of the Hamiltonian eigenstates, we can write:

$$\hat{H}[\boldsymbol{\mathcal{R}}(t)]\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle = \varepsilon_n[\boldsymbol{\mathcal{R}}(t)]\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle\,, \tag{1}$$

which allows us to determine the basis $|n[\boldsymbol{\mathcal{R}}(t)]\rangle$, but only up to a phase factor.

The evolution of parameters $\boldsymbol{\mathcal{R}}(t)$ is assumed to be adiabatic, meaning that a system initially in one of its eigenstates $|\psi_n(t=0)\rangle = |n[\boldsymbol{\mathcal{R}}(t=0)]\rangle$ remains in that eigenstate throughout the entire process, while accumulating a phase [10]:

$$|\psi_n(t)\rangle = C_n(t)\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle\,. \tag{2}$$

An intuitive example for such a process is a particle in a one-dimensional box, initially in its n-th excited state (i.e. $n+1$ nodes in the wavefunction). In an adiabatic process, if the walls of the box are moved slowly enough to the left or to the right, the particle remains in the same excited state and its wavefunction retains the same shape. Conversely, if the walls are moved non-adiabatically, the particle can end up in a superpositon of multiple states.

The arbitrariness of Equation (1) can be removed by making a phase choice, also known as a gauge, such that Equation (2) becomes:

$$|\psi_n(t)\rangle = e^{i\gamma_n(t)} e^{-i\int_0^t dt' \varepsilon_n[\boldsymbol{\mathcal{R}}(t')]}\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle\,. \tag{3}$$

The second exponential is known as the dynamical phase factor, while $\gamma_n(t)$ is the Berry phase and accounts for a special aspect of the state evolution which is not captured by the other factor [11].

Throughout this paper, the parameters our Hamiltonian depends on are the momentum vectors $\boldsymbol{k}$, and the parameter space is the Brillouin zone.

Plugging our state into the Schrödinger Equation $i\partial_t|\psi_n(t)\rangle = \hat{H}[\boldsymbol{k}(t)]|\psi_n(t)\rangle$ and taking the inner product with $\langle\psi_n(t)|$ on both sides, we get an expression for the Berry phase:

$$\frac{\partial}{\partial t}\gamma_n(t) = i\,\langle n[\boldsymbol{k}(t)]|\,\frac{\partial}{\partial t}\,|n[\boldsymbol{k}(t)]\rangle\,. \tag{4}$$

When evaluating the time derivative of the basis eigenvectors, one can write the Berry phase in terms of a new quantity $\boldsymbol{\mathcal{A}}^n(\boldsymbol{k})$, called the Berry connection or the Berry vector potential:

$$\begin{aligned}\gamma_n(t) &= \int_{\mathcal{C}} d\boldsymbol{k}\cdot i\,\langle n(\boldsymbol{k})|\,\nabla_{\boldsymbol{k}}\,|n(\boldsymbol{k})\rangle \\ &= \int_{\mathcal{C}} d\boldsymbol{k}\cdot\boldsymbol{\mathcal{A}}^n(\boldsymbol{k}),\end{aligned} \tag{5}$$

where $\mathcal{C}$ is now a path in the BZ.

However, from the way we defined Berry connection it can clearly be seen that it is gauge-dependent, meaning that we can always find a suitable choice of gauge to cancel out the phase γ_n accumulated on any open path.

This phase was neglected in the theoretical treatment of time-dependent problems for the most part until Michael Berry [12] first considered an adiabatic process where the motion was over a closed loop in parameter space. In this case he showed that the gauge-dependence

disappears, making Berry phase observable and having physical consequences (when it is non-vanishing). Also notable is the fact that γ_n does not depend explicitly on time, but rather on the path that $\mathcal{R}$ (or in our case, $\boldsymbol{k}$) traverses, which emphasises the geometric origin of the phase [13].

The analogy with electrodynamics should be fairly evident: $\boldsymbol{\mathcal{A}}^n \rightarrow$ vector potential of a magnetic field and $\gamma_n \rightarrow$ magnetic flux passing through a loop (in parameter space). For this analogy to be complete, the last concept we need to introduce is the Berry curvature Ω^n. In three dimensions, it corresponds to the magnetic field from the Berry vector potential:

$$\boldsymbol{\Omega}^n(\boldsymbol{k}) = \nabla_{\boldsymbol{k}} \times \boldsymbol{\mathcal{A}}^n(\boldsymbol{k}), \tag{6}$$

while for the special case of a two-dimensional parameter space, it takes the form [14]:

$$\begin{aligned}\Omega^n(\boldsymbol{k}) &= \partial_{k_x}\mathcal{A}_y(\boldsymbol{k}) - \partial_{k_y}\mathcal{A}_x(\boldsymbol{k}) \\ &= -2\,\mathrm{Im}[\langle\partial_{k_x} n(\boldsymbol{k})|\partial_{k_y} n(\boldsymbol{k})\rangle].\end{aligned} \tag{7}$$

Despite being written in a very compact form, Equation (7) may give rise to various problems if computed numerically, since the eigenstates $|n(\boldsymbol{k})\rangle$ are not gauge-invariant. At particular stages of the numerical analysis, the computer might attribute different arbitrary phases to these eigenstates, which are likely to produce errors in the final results.

A way around this is to use the fact that $|n(\boldsymbol{k})\rangle$ are eigenstates of the Hamiltonian and form a complete basis, which can help us rewrite Equation (7) as:

$$\Omega^n(\boldsymbol{k}) = -2\,\mathrm{Im}\sum_{n'\neq n} \frac{\langle n|\,\partial_{k_x} H\,|n'\rangle\,\langle n'|\,\partial_{k_y} H\,|n\rangle}{(\varepsilon_n - \varepsilon_{n'})^2}, \tag{8}$$

where we suppressed the $\boldsymbol{k}$ dependence to shorten our notation. This is now gauge-invariant and can be used for numerical computations.

This all links back to our initial discussion about topological invariants via Chern's theorem, which states that the integral of the Berry curvature over a closed surface is quantised in units of 2π. Since our parameter space is in fact the Brillouin Zone, we can write this as:

$$\int_{BZ} d\boldsymbol{k}^2 \cdot \boldsymbol{\Omega}^n(\boldsymbol{k}) = 2\pi C_1, \tag{9}$$

where C_1 is the first Chern number, as mentioned before.

C. Symmetry Considerations

There are two symmetry operations which we need to take into account before diving any further: space-inversion $\mathcal{I}$ and time-reversal $\mathcal{T}$. Their actions are:

$$\begin{cases} \mathcal{I} : \boldsymbol{r} \rightarrow -\boldsymbol{r} \\ \mathcal{I} : \boldsymbol{k} \rightarrow -\boldsymbol{k} \\ \mathcal{I} : t \rightarrow t \end{cases} \quad \text{and} \quad \begin{cases} \mathcal{T} : \boldsymbol{r} \rightarrow \boldsymbol{r} \\ \mathcal{T} : \boldsymbol{k} \rightarrow -\boldsymbol{k} \\ \mathcal{T} : t \rightarrow -t \end{cases}$$

Following these operations, it can be shown that under space-inversion symmetry Berry curvature is even (10), while under time-reversal symmetry it is odd (11):

$$\mathcal{I} : \boldsymbol{\Omega}_n(\mathbf{k}) \rightarrow \boldsymbol{\Omega}_n(-\mathbf{k}) = \boldsymbol{\Omega}_n(\mathbf{k}), \tag{10}$$

$$\mathcal{T} : \boldsymbol{\Omega}_n(\mathbf{k}) \rightarrow \boldsymbol{\Omega}_n(-\mathbf{k}) = -\boldsymbol{\Omega}_n(\mathbf{k}). \tag{11}$$

If both symmetries are present in a certain system, the Berry curvature must be zero in the entirety of the Brillouin zone, meaning that the Chern number is also zero, which corresponds to a topologically trivial phase. However, referring back to Equation (9), it is also important to note that $\boldsymbol{\Omega}_n(\mathbf{k}) \neq 0$ locally does not guarantee a topological phase either.

Breaking either of these two symmetries is not a continuous deformations of the Hamiltonian and therefore results in a gap opening in the band structure.

D. Quantum Anomalous Hall Effect

As already mentioned, Chern numbers were first mentioned in the context of condensed matter physics when studying the Quantum Hall Effect [9], so we will briefly discuss their importance in the case of the Quantum Anomalous Hall Effect.

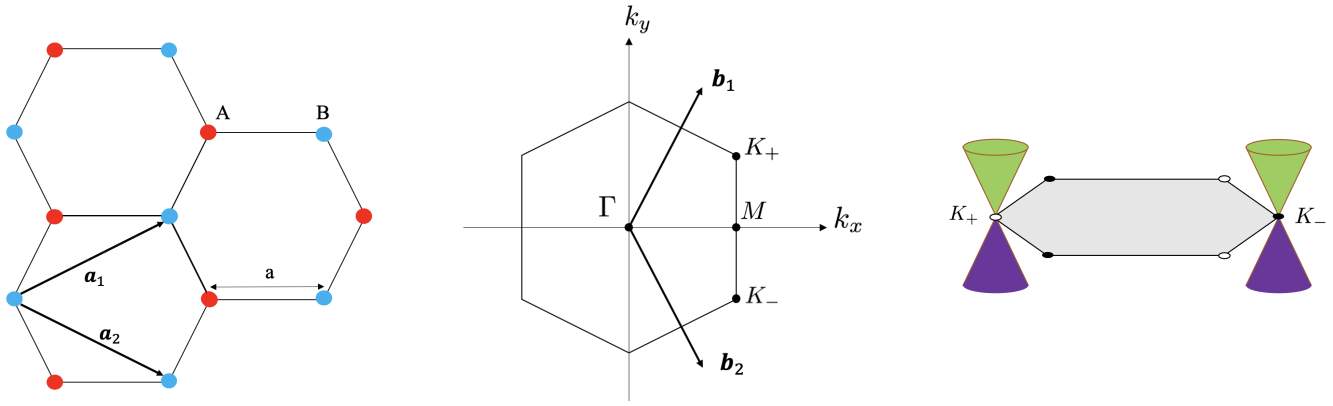
Similar to the Hall effect discovered by Edwin Hall in 1879 [15] which gives rise to a transverse current when a magnetic field is applied to a metal, its quantum counterpart was observed experimentally in 1980 by Klaus von Klitzing. In his paper "The quantized Hall effect" [16], he showed that in a two-dimensional electron system the Hall resistance is in fact quantised in units of e^2/h, which are natural constants that do not depend on the material or the sample used.

Unlike these two, the Quantum Anomalous Hall Effect (QAHE) is realised in the absence of a magnetic field. So far, it has been experimentally realised only by magnetically doping topological insulators [17] and in moiré heterostructures [18].

In magnetically doped topological insulators, the QAHE does not arise only due to the magnetic dopants which are responsible for breaking time-reversal symmetry, but also due to other topological properties, which emerge from spin–orbit coupling [19]. These two contributions lead to topologically non-trivial Chern bands, which are also present in moiré heterostructures where the Berry curvature gives rise to *orbital* instead of *spin* magnetic moments, which are then spontaneously polarised [20].

E. Monolayer and Bilayer Graphene

Graphene is made up of a triangular Bravais lattice with two basis atoms, labelled A and B, which gives the familiar honeycomb structure and a corresponding



(a) (b) (c)

FIG. 1: (a) Honeycomb structure of graphene, showing the two inequivalent sites A and B and lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. (b) Brillouin zone of graphene with high symmetry points Γ, M, K_+ and K_- and reciprocal lattice vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ (c) Inequivalent Dirac cones in reciprocal space, where the colours indicate valence and conduction bands.

hexagonal Brillouin zone (BZ), as shown in Figure 1(a) and 1(b). Since the two sites, A and B, cannot be connected by a lattice vector, they are non-equivalent. The low energy band structure, present around the high-symmetry points or valleys K_+ and K_-, can be calculated from a 2D massless Dirac Hamiltonian,

$$H = v_F \begin{pmatrix} \boldsymbol{\sigma} \cdot \mathbf{p}_{K_+} & 0 \\ 0 & -\boldsymbol{\sigma}^\dagger \cdot \mathbf{p}_{K_-} \end{pmatrix} , \qquad (12)$$

where v_F is the band velocity, $\boldsymbol{\sigma} = (\sigma_x, \sigma_y)$ is a vector of Pauli matrices and $\mathbf{p}_{K_+} = \hbar(\mathbf{k} - \mathbf{K}_+)$ and $\mathbf{p}_{K_-} = \hbar(\mathbf{k} - \mathbf{K}_-)$ are momenta measured with respect to the high-symmetry points. This Hamiltonian can be directly derived from a nearest-neighbour hopping tight-binding model through expansion around the K_+ and K_- points [21]. Note that H simultaneously describes both decoupled valleys - alternatively, it is common to write separate Hamiltonians for each valley. When solved, it gives a pair of non-equivalent Dirac cones, characterised by a linear dispersion, whose apexes meet at the Dirac point - also referred to as the charge neutrality point (CNP). This is shown in Figure 1(c).

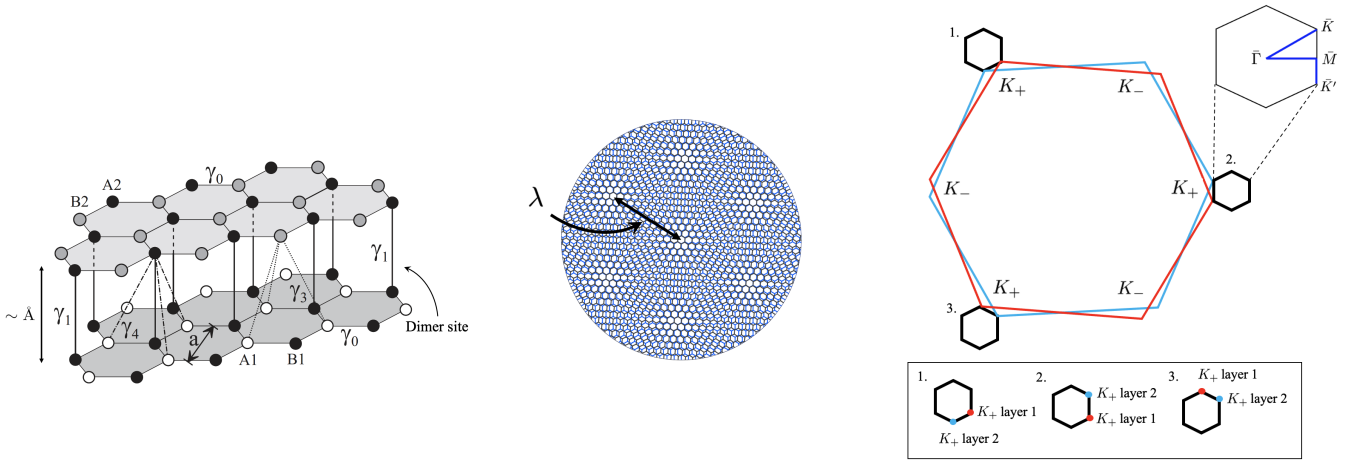
The multi-component nature of the wavefunction is not related to the true electron spin, as is the case with the Dirac equation, but is a pseudo-spin property which results from the valley degeneracy. We thus have a four-fold band degeneracy: two from the valley degree of freedom and two from spin. The pseudospin variable gives electrons in the K_+ and K_- valleys opposite chirality.

The band-touching points are protected by time-reversal and space-inversion symmetries, which must be broken to open up a gap. This can be achieved, for example, through the introduction of a staggered potential (inversion symmetry breaking) or application of a perpendicular magnetic field (time-reversal symmetry breaking). The former is present in hexagonal boron nitride (hBN), which shares the same honeycomb structure as graphene, but with two alternating atomic species.

Bilayer graphene (BG) - made up of two coupled monolayers - is most commonly stacked in the *Bernal* or AB configuration, as shown in Figure 2(a). Here, there are four basis atoms in the unit cell: A1, B1 in the bottom layer and A2, B2 in the top layer. Atom B1 sits directly below A2 - this is known as a dimer site, since interlayer coupling is strong. Atoms A1 and B2 lie at the centre of each opposing layers hexagon and are referred to as non-dimer sites. This is the most energetically stable configuration, and is how graphite is stacked [22]. Similarly, AA stacking is when the two graphene sheets lie directly on top of each other.

The γ_i in Figure 2(a) are hopping parameters in a tight-binding model for BG, representing the probability of an electron migrating from one site to another. γ_0 is the intralayer coupling parameter present in monolayer graphene; γ_1 gives the dimer site interlayer coupling; γ_3 couples non-dimer orbital pairs A1-B2, resulting in trigonal warping of the band structure; and γ_4 is interlayer coupling between A1-A2, B1-B2 pairs and gives electron-hole asymmetry. These can be determined by fitting tight-binding calculations with experimental measurements as obtained in Raman [23] and infrared spectroscopy [24], or through the use of first principles density functional theory calculations [25].

As in monolayer graphene, a tight-binding model can be used to construct an effective continuum model valid at low energies for BG. This gives a parabolic dispersion at low momenta and thus a non-zero electron mass. While the monolayer continuum Hamiltonian produced massless chiral particles, in bilayer graphene we find massive chiral particles.



(a) (b) (c)

FIG. 2: (a) Structure of AB stacked bilayer graphene. A dimer site is labelled, where an atom from the top layer of the A sublattice sits directly above from an atom of the B sublattice of the bottom layer. The hopping parameters $\gamma_1, \gamma_2, \gamma_3, \gamma_4$ are shown. Taken from [26]. (b) The moiré pattern created by two hexagonal lattices with a twist angle between them. The rough periodicity λ of the structure is labelled. (c) In red and blue are the BZ's of the monolayers, twisted by some angle θ. In black are the mini moiré BZ, whose side lengths are spanned by the shift between corners. The set of high symmetry points in the moiré BZ, labelled with a bar notation, are shown. Three different hopping processes can occur between the equivalent K_+ points of the two layers.

F. Moiré Theory: Twisted Bilayer Graphene

Moiré patterns are frequently observed in the macro and microscopic world: they require only two identical or similar superimposed meshes to be created. Beyond their aesthetic and mathematical properties, they were first discovered - by Li *et. al* [27] - to have significant effects on the electronic properties of twisted bilayer graphene (tBG). For this reason, tBG was responsible for the inception of a field now referred to as *twistronics* and as such, is the basis of other, more complex 2D moiré systems.

Figure 2(b) shows the moiré pattern created by superposing two honeycomb lattices with a relative twist angle, θ. The label λ is the moiré wavelength, indicating the scale at which the pattern is roughly periodic. For a generic twist angle, the resulting structure is not rigorously periodic and Bloch's theorem does not apply, making it impossible to carry out band structure calculations. Only for commensurate angles, where the moiré lattice vector can be written as an integer sum of the underlying lattice vectors, is there true periodicity. However, in the small angle regime ($\theta \lesssim 10°$), where $\lambda \gg a$, and a is the lattice parameter, the structure becomes *approximately* periodic and it can be shown [28, 29] that Bloch's theorem applies even for incommensurate angles. In this regime, the moiré period goes as $\lambda = a/[2\sin(\theta/2)]$ [29].

The large hexagons in Figure 2(c) are the first BZ's of the two monolayer graphene sheets, while the smaller hexagons are the mini moiré BZ's (MBZs), with a side length spanned by the shift between corners. The high symmetry corners in this mini BZ will be referred to as $\bar{K}$ and $\bar{K}'$.

The electrons which are important - those responsible for the low energy features - will have Bloch waves which sit at the corners K_- and K_+ of the monolayer BZ's. When the electrons of the two layers interact at these regions, the three equivalent Dirac points lead to three different hopping processes, since the orientations of the Bloch waves are different in each case. This is shown in the three MBZs of Figure 2(c). Conservation of momentum must hold, and so a superposition of these hopping processes will capture the moiré interference.

Lopes dos Santos *et. al* [30] were the first to theoretically examine the effect of twisting BG on its electronic structure. They used a continuum model, with a slowly varying periodic interlayer coupling to capture the effects introduced by the moiré pattern. For a small commensurate twist angle they found a linear energy dispersion, as in the monolayer case. In 2011, Bistritzer and Macdonald [28] elaborated on this model, and derived a set of discrete magic angles for which the lowest bands flatten. These works formed the basis for much of the subsequent theoretical studies of moiré systems.

The continuum Hamiltonian of tBG can be written as the sum $H_{\text{cont}} = H_{\text{Dirac}} + H_T$, where H_T describes the tunnelling effects introduced by the moiré interference. If we rotate each layer by an angle $\pm\theta/2$, the continuum

Hamiltonian for the K_+ valley can be written [31],

$$H_{K_+} = \begin{pmatrix} v_F \boldsymbol{\sigma}_{\theta/2} \cdot \mathbf{p}_+^{(1)} & T_m(\mathbf{r}) \\ T_m^\dagger(\mathbf{r}) & v_F \boldsymbol{\sigma}_{-\theta/2} \cdot \mathbf{p}_+^{(2)} \end{pmatrix} , \tag{13}$$

where $\mathbf{p}_+^{(l)} = \hbar(\mathbf{k} - \mathbf{K}_+^{(l)})$, $l = 1, 2$, are the momenta for the top and bottom layer respectively, $\boldsymbol{\sigma}_{\theta/2} = e^{-i\frac{\theta}{4}\sigma_z}(\sigma_x, \sigma_y)e^{i\frac{\theta}{4}\sigma_z}$ are the rotated Pauli matrices and $T_m(\mathbf{r})$ is the interlayer moiré potential which couples the two layers. It can be expressed as,

$$T_m(\mathbf{r}) = \sum_{j=1}^{3} T_j \exp(-i\mathbf{q}_j \cdot \mathbf{r}) , \tag{14}$$

where $\mathbf{q}_j$ are the momentum transfers of the three distinct hopping process described previously and,

$$T_{j+1} = w_0\sigma_0 + w_1(\cos(\phi_j)\sigma_x + \sin(\phi_j)\sigma_y) , \tag{15}$$

where $\phi = 2\pi/3$, σ_0 is the identity and w_0, w_1 are hopping energies for AA and AB coupling respectively. The Hamiltonian for the K_- valley can be obtained by applying a time-reversal operation.

G. Hofstadter's Butterfly

Electrons confined to two dimensions in a magnetic field follow cyclotron orbits and coalesce into highly degenerate Landau levels. Under a periodic lattice potential, electrons occupy quasi-continuous Bloch bands. When the effect of both potentials is present simultaneously, the resulting energy structure - shown in Figure 3 - is considerably more complex, as was initially predicted by Hofstadter [32] for a 2D square lattice. This self-similar, fractal structure known as Hofstadter's butterfly (HB) is a consequence of two competing length scales: the lattice (a) and magnetic ($l_B = \sqrt{\hbar/eB}$) periodicity. When these are of the same order, fractal patterns can be observed. For a typical crystal lattice with a period of a few Angstroms, experimentally unfeasible magnetic fields as large as 10,000T are required to meet this condition. Through lithographic means however, artificial superlattices in GaAs/AlGaAs heterostructures [33, 34] lowered the magnetic field threshold and allowed experimental fingerprints of Hofstadter's spectrum to be observed. These were lacking in clarity as a result of disorder introduced in the superlattice fabrication process. The natural, long-range periodicity in moiré systems served as a superior platform for observing the fractal spectrum. Using bilayer graphene on hBN, Dean *et. al* [35] obtained clear evidence of the recursive energy spectrum. Working with similar van der Waals heterostructures, other groups [36, 37] have also reproduced the HB in moiré systems.

It is possible to express the information within a HB spectrum in linear trends known as Wannier diagrams.

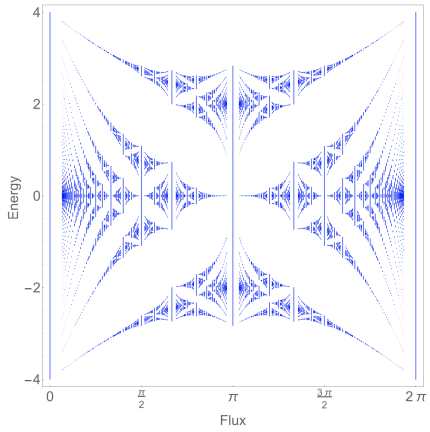


FIG. 3: Hoftadter's butterfly spectrum - the Landau level structure of a two dimensional square lattice under the influence of a perpendicular magnetic field. Between the minigaps, a complex self-repeating structure is observed. Taken from [38].

Plotting the integrated charge-carrier density as function of flux gives the Diophantine relation,

$$n/n_0 = \Phi/\Phi_0 t + s , \tag{16}$$

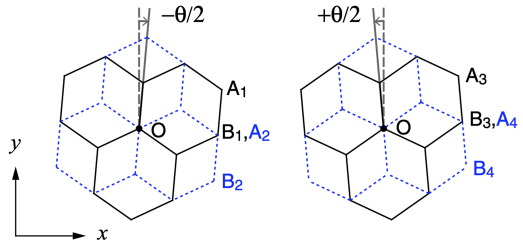
where n/n_0 and Φ/Φ_0 are the normalised carrier density and flux respectively. The integer t is the Chern number which appears in the quantised longitudinal conductivity $\sigma_{xy} = te^2/h$ and labels each minigap in the spectrum, while s represents the Bloch band filling index.

III. TWISTED DOUBLE BILAYER GRAPHENE

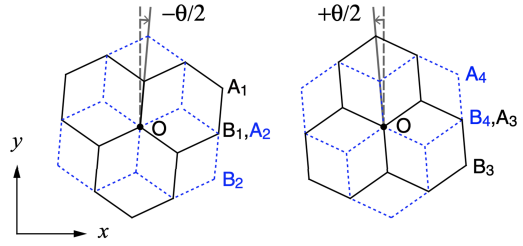
The recent experimental observation of flat bands, unconventional ferromagnetic superconductivity and correlated insulating states in twisted *double* bilayer graphene (tDBG) [39–41] - as was observed in tBG - calls for further theoretical and practical efforts to elucidate upon its properties. In tBG, the flat bands understood to be responsible for the correlated behaviour have been shown to host fragile topology [42] - a unique topological state characterised by odd windings of Wilson loops [43]. The topologically non-trivial bands have been proposed to originate from two zeroth pseudo LL's, which have opposite sublattice polarizations [44]. These developments in tBG have motivated work in tDBG to explore its topological properties. A brief summary of the current progress in this direction will be provided.

A. Atomic Structure

Two distinct stacking configurations for tDBG are commonly considered in the literature, known as AB-AB and AB-BA. These apply the same conventions as described previously for BG and are shown in Figure 4(a)



(a) AB-AB



(b) AB-BA

FIG. 4: The structure of tDBG for (a) AB-AB and (b) AB-BA. Each bilayer is rotated by an angle $\pm\theta/2$ and has four basis atoms. Adapted from [45].

and 4(b). The AB-AB structure has two Bernal bilayers, while in AB-BA the second bilayer is rotated by 180°. Here, a unit cell is made up of eight atoms: A1, B1, A2, B2 in the first bilayer and A3, B3, A4, B4 in the second bilayer. These are rotated by an angle $+\theta/2$ and $-\theta/2$ respectively, giving a total twist angle of θ. Both stacking configurations have C_{3z} symmetry - 3-fold rotation about the z axis - but differ in their two-fold rotational symmetry, with AB-AB and AB-BA having C_{2x} and C_{2y} symmetries respectively.

The lattice vectors of each bilayer can be written,

$$\mathbf{a}_1 = a(1,0)\,, \qquad \mathbf{a}_2 = a(1/2,\sqrt{3})\,, \tag{17}$$

where $a \approx 2.46$ Å is the lattice parameter of graphene. The reciprocal lattice vectors are then,

$$\mathbf{b}_1 = \frac{2\pi}{a}\left(1,-\frac{1}{\sqrt{3}}\right), \qquad \mathbf{b}_2 = \frac{2\pi}{a}\left(0,\frac{2}{\sqrt{3}}\right). \tag{18}$$

When the bilayers are each rotated by an angle $\mp\theta/2$, the lattice and reciprocal lattice vectors of the lth bilayer, $l = 1, 2$, become,

$$\mathbf{a}_i^{(l)} = R(\mp\theta/2)\mathbf{a}_i\,, \qquad \mathbf{b}_i^{(l)} = R(\mp\theta/2)\mathbf{b}_i\,. \tag{19}$$

where R is a rotation matrix. In the small angle regime, where the distinction between commensurate and incommensurate cases can be ignored, the moiré reciprocal lattice vectors are,

$$\mathbf{G}_i^M = \mathbf{b}_i^{(1)} - \mathbf{b}_i^{(2)} \tag{20}$$

for $i = 1, 2$. The Dirac points K_+ and K_- of each graphene sheet are given by,

$$\mathbf{K}_\xi^{(l)} = \frac{\xi}{3}(2\mathbf{b}_1^{(l)} + \mathbf{b}_2^{(l)}) \tag{21}$$

where $\xi = \pm 1$ is the valley index.

B. Models

Tight-binding and continuum models are the most commonly implemented approaches when describing moiré systems. They have been shown to give equivalent band structures for tBG [46]. A continuum method used by Koshino [45], based on the Dirac Hamiltonian will be presented here. Taking into account one $2p_z$ orbital for each basis atom, the tDBG Hamiltonian for the AB-AB structure and valley $\xi = \pm 1$ is,

$$H_{\text{AB-AB}} = \begin{pmatrix} H_0(\mathbf{k}_1) & g^\dagger(\mathbf{k}_1) & & \\ g(\mathbf{k}_1) & H_0'(\mathbf{k}_1) & T_m^\dagger & \\ & T_m & H_0(\mathbf{k}_2) & g^\dagger(\mathbf{k}_2) \\ & & g(\mathbf{k}_2) & H_0'(\mathbf{k}_2) \end{pmatrix} + V\,. \tag{22}$$

Here $\mathbf{k}_l = R(\pm\theta/2)(\mathbf{k} - \mathbf{K}_\xi^{(l)})$ is the Bloch wave-vector for the moiré lattice centered at valley sites with $\pm$ for $l = 1, 2$. $H_0(\mathbf{k})$ and $H_0'(\mathbf{k})$ are Dirac Hamiltonians,

$$H_0(\mathbf{k}) = -\hbar v \begin{pmatrix} 0 & \xi k_x - i k_y \\ \xi k_x + i k_y & \Delta' \end{pmatrix}\,, \tag{23}$$

$$H_0'(\mathbf{k}) = -\hbar v \begin{pmatrix} \Delta' & \xi k_x - i k_y \\ \xi k_x + i k_y & 0 \end{pmatrix}\,, \tag{24}$$

but with an added potential Δ' which describes the energy difference between dimer and non-dimer sites. The g matrices describe interlayer coupling within bilayers,

$$g(\mathbf{k}) = \begin{pmatrix} \hbar v_4(\xi k_x + i k_y) & \gamma_1 \\ \hbar v_3(\xi k_x - i k_y) & \hbar v_4(\xi k_x + i k_y) \end{pmatrix}\,, \tag{25}$$

where $v_i = (\sqrt{3}/2)\gamma_i a/\hbar$ $(i = 3, 4)$, with γ_i being the hopping parameters described for BG. The values for parameters $\Delta', \gamma_1, \gamma_3, \gamma_4$ were determined experimentally through comparison with the Slonczewski-Weiss-McClure model for bulk graphite [22, 47]. The moiré interlayer coupling has three distinct contributions,

$$U = \begin{pmatrix} u & u' \\ u' & u \end{pmatrix} + \begin{pmatrix} u & u'\omega^{-\xi} \\ u'\omega^{\xi} & u \end{pmatrix} e^{i\xi \mathbf{G}_1^M \cdot \mathbf{r}} + \begin{pmatrix} u & u'\omega^{\xi} \\ u'\omega^{-\xi} & u \end{pmatrix} \times e^{i\xi(\mathbf{G}_1^M + \mathbf{G}_2^M)\cdot \mathbf{r}}\,, \tag{26}$$

where $\omega = e^{2\pi/3}$ and u and u' are hopping energies often set to be equal in earlier moiré continuum models [26, 28, 48]. When $u = u'$, the separation between bilayers is constant, such that the structure is flat. However, it has been theoretically argued [49–51] that a corrugation

effect exists in tBG which causes AA regions in the moiré pattern to have have greater interlayer spacing than AB or BA regions. Koshino *et al.* [52] derived different values $u = 0.0797$ eV and $u' = 0.0975$ eV to model the lattice relaxation in tBG, which are used in the model for Equation 22. The effect of this on the band structure is to increase the energy gap between the lowest and excited bands, in agreement with experimental observations [6, 7]. The magnitude of this effect in tDBG has not been derived, but is expected to be lower than in tBG due to its increased stiffness. Haddadi *et al.* [53] argue that controlling the u/u' ratio is not sufficient to model the lattice relaxation, and that the hopping parameters γ_3 and γ_4 must be renormalized from their values in tBG to describe tDBG.

The final term in Equation 22 is the interlayer asymmetric potential, which acts as a perpendicular electric field and is chosen to be constant,

$$V = \begin{pmatrix} \frac{3}{2}\Delta\hat{1} & & & \\ & \frac{1}{2}\Delta\hat{1} & & \\ & & -\frac{1}{2}\Delta\hat{1} & \\ & & & -\frac{3}{2}\Delta\hat{1} \end{pmatrix} , \tag{27}$$

where Δ is the potential energy between layers.

The Hamiltonian for the AB-BA configuration has the same structure, except $H_0(\mathbf{k}_2)$, $H_0'(\mathbf{k}_2)$ and $g(\mathbf{k}_2)$, $g^\dagger(\mathbf{k}_2)$ switch places in the bottom right quadrant,

$$H_{\text{AB-BA}} = \begin{pmatrix} H_0(\mathbf{k}_1) & g^\dagger(\mathbf{k}_1) & & \\ g(\mathbf{k}_1) & H_0'(\mathbf{k}_1) & T_m^\dagger & \\ & T_m & H_0'(\mathbf{k}_2) & g(\mathbf{k}_2) \\ & & g^\dagger(\mathbf{k}_2) & H_0(\mathbf{k}_2) \end{pmatrix} + V . \tag{28}$$

The symmetries of the system can be examined to derive constraints on the Berry curvature, which then inform the topological properties. The Hamiltonian $H_{\text{AB-AB}}$ for each valley commutes with C_{2x} when $\Delta = 0$. As such, we expect the Berry curvature to invariant under C_{2x} which gives the constraint,

$$\Omega_{n,(k_x,-k_y)} = -\Omega_{n,(k_x,k_y)} , \tag{29}$$

so that for any band n of the AB-AB structure, the Chern number is expected to be zero. In the case of AB-BA, an application of C_{2y} exchanges valleys $\xi = \pm$ so that $H_{\text{AB-BA}}$ only commutes with $C_{2y}\mathcal{T}$ when $\Delta = 0$. This imposes the condition,

$$\Omega_{n,(k_x,-k_y)} = \Omega_{n,(k_x,k_y)} , \tag{30}$$

which means that for any band n, the Chern number is allowed to be finite.

To gauge the effect of parameters γ_3, γ_4 and Δ' on the electronic structure, it is common [45, 53, 54] to define a *minimal* model in which these are neglected. This has the effect of ensuring electron-hole symmetry in the energy bands $E_{n,\mathbf{k}} = -E_{-n,-\mathbf{k}}$ for both stacking configurations, but anti-symmetry $C_{-n} = -C_n$ in the Chern number for AB-AB and electron-hole symmetry $C_{-n} = C_n$ in the Chern number for AB-BA.

C. Electronic and Topological Properties

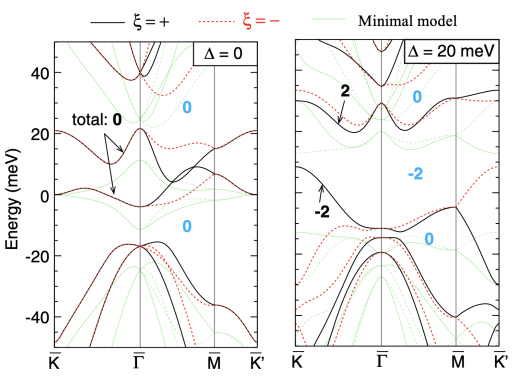
Figure 5 shows the band structures of a) AB-AB and b) AB-BA tDBG configurations for a twist angle $\theta = 1.33°$ through the path $\bar{K}$-$\bar{\Gamma}$-$\bar{M}$-$\bar{K}'$ in the MBZ, obtained with the effective continuum model. Results for both $\Delta = 0$ eV and $\Delta = 20$ meV are shown. In black and dashed red, are the bands obtained for the $\xi = \pm$ valleys respectively, using the full parameter model (Equations 22 and 28), while the green lines show the minimal model. In black are the Chern numbers of the labelled bands for the $\xi = +$ valley. Due to time reversal symmetry, the $\xi = -$ valley Chern numbers are opposite in sign. The blue numbers are the result of summing all of the valley Chern numbers up to an energy gap.

Firstly, note that this system - as opposed to tBG - does not have inversion symmetry [55], and as such, it is expected a gap will open in the spectrum at the CNP. These are present even when $\Delta = 0$ for both structures. The central bands (valence and conduction bands) are not symmetric in either case, as is present in the minimal model. As expected from the symmetry considerations, the Chern numbers for all bands are vanishing in AB-AB when $\Delta = 0$, but are finite for some bands in AB-BA. Thus, it is possible to label AB-AB as a trivial insulator in the absence of a perpendicular electric field, while AB-BA is a valley Hall insulator if the Fermi energy lies in a gap.

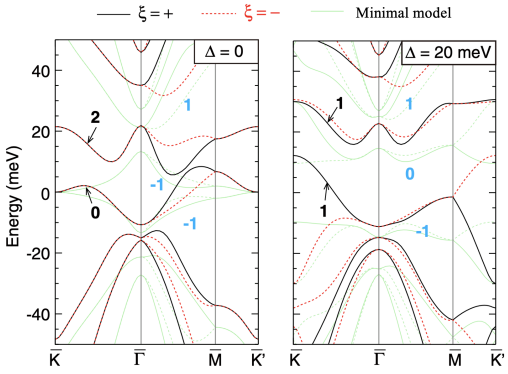
When Δ is turned on, the gap between the lowest energy bands increases and the bandwidth is generally found to decrease. In AB-AB, the band touching points present along the Γ-M line that were protected by C_{2x} symmetry are gapped out, resulting in a topological phase transition and finite Chern numbers. For AB-BA, the valence and conduction bands touch between $\Delta = 0$ and $\Delta = 20$ meV, so that the Chern number is transferred between them - from (2, 0) to (1, 1).

Chebrolu *et al.* [54] also use a continuum model to describe the electronic properties of tDBG. The band structures obtained with a minimal and full parameter model for a twist angle of $\theta = 1.5°$ and $\Delta = 30$ meV are shown in Figure 6 for (a) AB-AB and (b) AB-BA. Only the K_+ valley bands are shown. These calculations are in agreement with those of Koshino: both band structures are gapped at the CNP and have non-zero Chern numbers. In particular, the sum of the Chern numbers of the valence and conduction bands are zero in AB-AB and non-zero in AB-BA. This is a consequence of the opposite or equal chiralities of the bands belonging to the top and bottom layers of the two structures.

Both Koshino [45] and Chebrolu *et al.* [54] find that, despite having different dispersions, the minimal and full parameter models have identical Chern numbers for the central bands. This demonstrates the robustness of the bands to perturbations introduced by the hopping parameters γ_3 and γ_4, which indicates that the topological

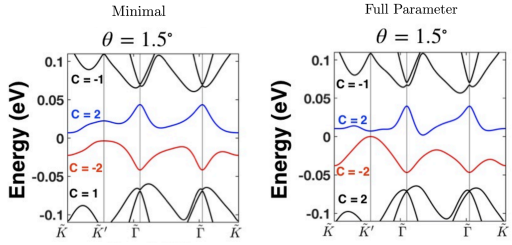


(a) AB-AB

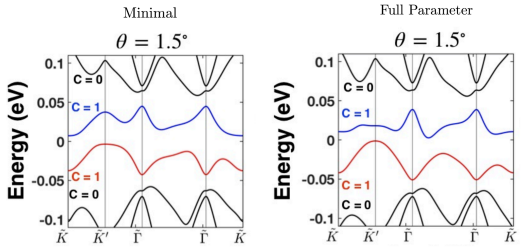


(b) AB-BA

FIG. 5: Band structure of tDBG through a path in the moiré BZ obtained by Koshino [45] using the continuum model with $\Delta = 0$ (left) and $\Delta = 20$ meV (right) for (a) AB-AB and (b) AB-BA. The black and dashed red lines are results for the two valleys $\xi = \pm$ respectively, and in green are those for the minimal model. Chern numbers for the $\xi = +$ bands are labelled in black, and in blue are $\xi = +$ Chern numbers summed over all bands below the energy gap. The corresponding ξ_- bands have opposite sign Chern numbers.



(a) AB-AB



(b) AB-BA

FIG. 6: Band structure of tDBG in the moiré BZ obtained by Chebrolu *et al.* [54] using the minimal and full parameter models at a twist angle $\theta = 1.5°$ for (a) AB-AB and (b) AB-BA. The red and blue bands are the valence and conduction bands respectively. Only the K_+ valley bands and their Chern numbers are shown.

properties may be derived with models that lack much of the higher order complexity present in real systems.

The band structures, particularly for finite Δ, are almost indistinguishable between the two stacking configurations, but topologically are drastically different. This highlights that the information about the topology is contained within the Bloch states and their distribution in momentum space, which cannot be expressed using only the eigenvalues of the Hamiltonian.

Naturally, this poses the problem of observing the topology through experiment. Wu *et al.* [56] theoretically explore the possibility of using the Landau level structure to probe the topological properties of tDBG. They use a tight binding model with lattice relaxation effects obtained from atomistic classical force field simulations. Applying a uniform magnetic field B perpendicular to the graphene planes, the energy bands are calculated perturbatively as,

$$\varepsilon_{n,\sigma,\xi}(\mathbf{k}, B) = \varepsilon_{n,\sigma}(\mathbf{k}) + \mu_B g \sigma B + m_{n,\xi}(\mathbf{k}) B \,, \qquad (31)$$

where n is the band index, $\sigma = \pm\frac{1}{2}$ for electron spin, $\xi = \pm 1$ is still the valley index, and μ_B is the Bohr magneton. The first term on the right hand side is the energy dispersion without perturbations, the second term accounts for the Zeeman effect, which is ignored by the authors, and the third term is an orbital magnetisation contribution given by,

$$m_{n,\xi}(\mathbf{k}) = -e\hbar \,\mathrm{Im} \sum_{l \neq n} \frac{\langle n, \xi \,|\partial_{k_x} \mathcal{H}_\xi| \, l, \xi\rangle \langle l, \xi \,|\partial_{k_y} \mathcal{H}_\xi| \, n, \xi\rangle}{\varepsilon_{n,\xi,\mathbf{k}} - \varepsilon_{l,\xi,\mathbf{k}}} \,, \qquad (32)$$

where $|n, \tau\rangle$ and $|l, \tau\rangle$ are eigenstates of the Hamiltonian $\mathcal{H}_\tau$. Figure 7 shows the HB spectra, Wannier diagrams and Hall conductivities obtained by Wu *et al.* for the AB-AB (a-c) and AB-BA (d-f) configurations. Brighter and darker regions indicate higher or lower density of states. The white numbers in the Landau levels plots are the Chern numbers of the gaps, or equivalently, the gradients of the Wannier diagram lines. Differences in the HB spectra between the two configurations are apparent - considerably more levels are present in AB-BA even in the small range of fields plotted. To see why this is, note that if C_{2x} symmetry is present,

$$m_n(k_x, k_y) = -m_n(k_x, -k_y) \,, \qquad (33)$$

such that the orbital magnetisation will be the same for the bilayer valleys K_+ and K_-, but opposite for the

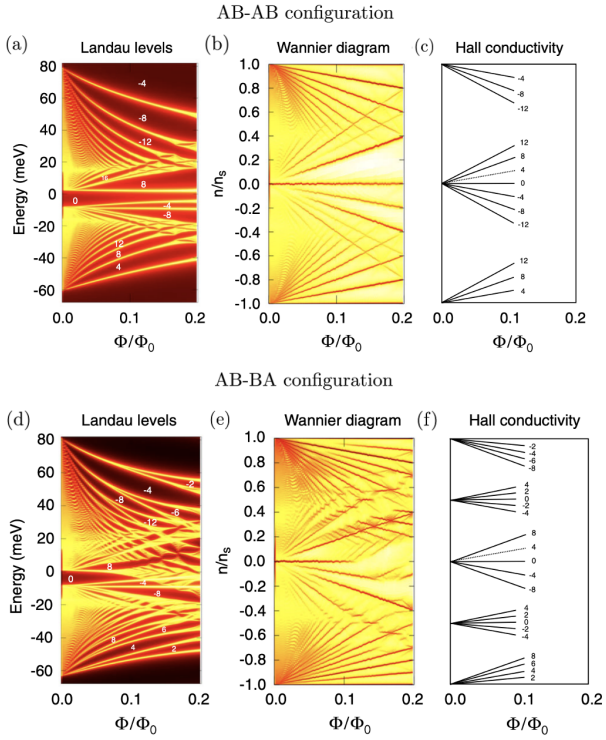


FIG. 7: Theoretical calculations of the HB spectra in tDBG from [56]. Landau levels, Wannier diagrams and Hall conductivity for (a)-(c) AB-AB and (d)-(f) AB-BA.

moiré valleys $\bar{K}$ and $\bar{K}'$. Alternatively, if C_{2y} symmetry is present,

$$m_n(k_x, k_y) = -m_n(-k_x, k_y) \ , \tag{34}$$

so that the converse is true: $m_{n,\xi}(\mathbf{k})$ will be the same for the moiré valleys $\bar{K}$ and $\bar{K}'$, but opposite for the bilayer valleys K_+ and K_-. From Equation 31, it is evident that bilayer valley degeneracy will be lifted if in the AB-BA configuration, since C_{2y} symmetry is present.

Differences in the Landau level structure between the two configurations are a result of the distribution of orbital magnetisation in momentum space, which is closely related to the Berry curvature. As such, the differences between AB-AB and AB-BA can be traced back to the topological properties.

D. Experimental Work

Although currently limited, a number of experimental studies probing the topological properties of tDBG are present in the literature [57–60]. Of these, the most pertinent is by He *et al.* [57], who explicitly compare the two stacking configurations with electrical transport measurements. Using the cut-and-stack approach, they fabricate five AB-AB and six AB-BA devices with a range of twist angles from 1.06° to 1.39°. A perpendicular electrical displacement field D and magnetic field B are applied. The number of electrons per moiré unit cell or filling factor ν, is controlled by varying the carrier density n. As anticipated from the similarities in the calculated band structures, He *et al.* find that the main transport features through a range of D and B fields are very similar for the two configurations.

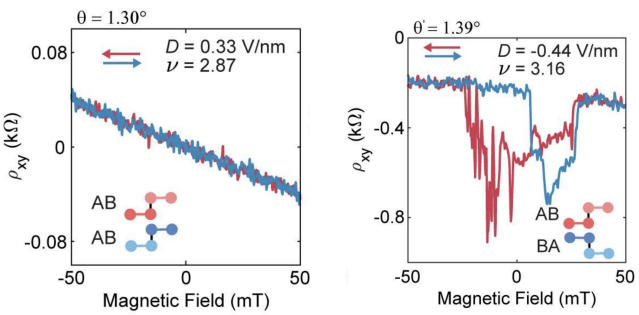
However, some subtle, but important distinctions were observed between the two arrangements. Figure 8(a) and 8(b) show the effect of sweeping the magnetic field back and forth on the Hall resistivity ρ_{xy} for AB-AB and AB-BA respectively, with band filling near $v = +3$. An anomalous Hall effect (AHE) - a non-zero Hall voltage without an external magnetic field - is implied by the hysteresis of Figure 8(b) at $B = 0$. This suggests spontaneous time-reversal symmetry breaking, which has important consequences on the Berry curvature and likewise, the topological properties. The same hysteresis is not observed in the AB-AB configuration, suggesting a preservation of time-reversal symmetry.

Further hints on the topological differences between the structures are provided by probing the evolution of ρ_{xy} at larger magnetic fields. This is shown in Figure 9(a) for AB-BA at different filling factors and magnetic fields. The quantised trajectories of the resistivity are linear and can be described with an expression analogous to the Diophantine relation, where the carrier density is replaced by ν, as shown Figure 8(b). Purple lines indicate integer quantum Hall states; red, orange and blue lines are Chern insulators with different intercepts and black lines are topologically trivial states. A unique state denoted by the dashed line, not observed in AB-AB, with a Chern number $C = 1$ and band filling index $s = 7/2$ is present in AB-BA. This is referred to as a symmetry broken Chern insulator by the authors, and its existence - again - is attributed to the distinct differences in the Berry curvature and valley Chern numbers of the two stacking configurations.

Evidently, the stacking *chirality* can offer a new degree of freedom in twisted multilayer graphene systems, allowing control over topological properties. This presents the potential to use topological changes in the electronic bands to differentiate between unknown samples with different stacking chiralities. Future work may reveal that this property is also present in a broader range of moiré systems and vdW heterostructures, further contributing to the rich landscape of emergent topological phenomena in this field.

IV. GRAPHENE/HBN SUPERLATTICES

The theoretical description underlying tBG has been well understood ever since the proposal of the Bistritzer and MacDonald model in 2011 [28] and has been probed experimentally continuously over the past decade. De-



(a) AB-AB (b) AB-BA

FIG. 8: Hall conductivity ρ_{xy} in tDBG as the magnetic field is swept forwards (blue) and backwards (red) for (a) AB-AB and (b) AB-BA. The twist angle θ, displacement field D and filling factor ν are shown for each case. Taken from [57].

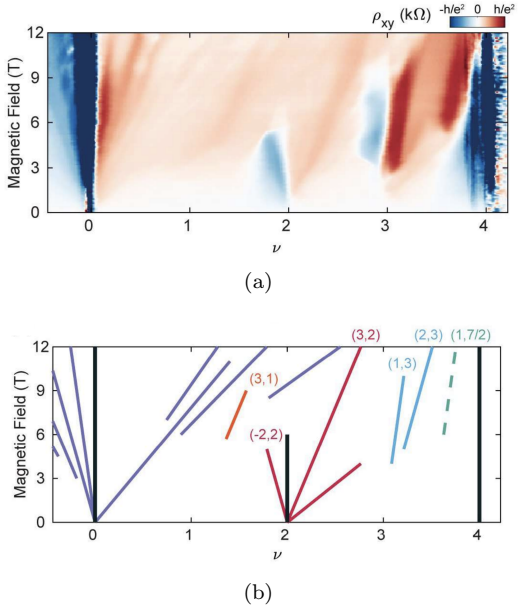


FIG. 9: (a) Hall conductivity ρ_{xy} in AB-BA tDBG as a function of magnetic field and filling factor - commonly referred to as a Landau fan diagram. (b) Schematic of the gapped states present in (a). The numbers (C, s) are the Chern number and band filling index. Colours indicate different values of s. Black lines have $C = 0$ and so are topologically trivial. The dashed line $(1, 7/2)$ is the unique symmetry broken Chern insulator state. Taken from [57].

spite the fact that experimental measurements on tBG are usually performed in samples either encapsulated between two hBN clapping layers or suspended on a hBN substrate [61], the theoretical description of tBG on hBN runs into a fundamental difficulty when the substrate is nearly aligned with the adjacent graphene layers.

Because of the small lattice parameter mismatch between graphene, $a_G \approx 2.46\,\text{Å}$, and hBN, $a_{BN} \approx 2.5\,\text{Å}$, a new moiré pattern is produced when the two layers are nearly aligned [62][63]. So far it was considered that only a spatially averaged sublattice energy difference is retained in tBG, giving rise to broken space-inversion symmetry [64]. However, a more detailed theoretical model considers an extra position-dependent tBG/hBN interaction term, which plays a crucial role when studying how commensurability can be related to topological properties of the system, like the QAHE.

A. General Description

This analysis focuses on a trilayer system: tBG sitting on a monolayer of hBN, where the graphene adjacent to hBN is labeled as G1 and it is considered fixed. The other two layers, G2 and hBN, are twisted relative to it with angles θ_{GG} and θ_{GBN}, as shown in Figure 10(a).

For particular combinations of the twist angles θ_{GG} and θ_{GBN}, the moiré patterns formed by each of the two layers are commensurate. When talking about commensurability of two layers, we refer to the fact that the moiré pattern they form has translational symmetry, which is emphasised in Figure 11(a) by the two moiré reciprocal vectors. However, if the twist angle is incommensurate, translational invarience is lost, which can be seen in Figure 11(b) by noting that the two highlighted hexagons are similar, but not exactly identical [66].

The commensurability of two moiré patterns can be understood in a similar way to the commensurability of two single layers, discussed above. Hence a system formed by two moiré patterns G1/G2 and G1/hBN which are each twisted at a commensurate pair of angles is periodic in a larger, translationally symmetric unit cell [65].

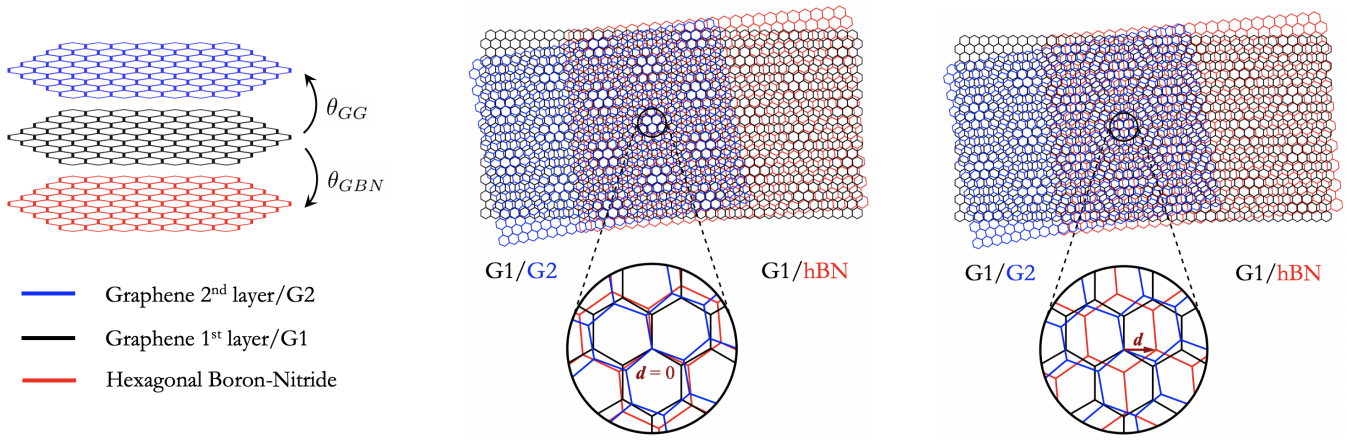
Unlike the previous section, our analysis will now focus on an AA stacking type in the graphene bilayer, which happens when the two graphene sheets lie directly on top of each other. We fix the local AA stacking point around which we twist the G2 layer as the origin of our system.

The hBN layer can shift from the origin by a rigid displacement $\boldsymbol{d}$, such that when $\boldsymbol{d} = 0$ we have a local AAA stacking, which happens when an atom of Carbon from G2 sits on top of a Carbon atom from G1, which in turn sits on a Boron atom from hBN. This is clearly shown in Figure 10(b), while in 10(c) we can see the new moiré pattern formed when hBN is shifted by $\boldsymbol{d} \neq 0$.

In a commensurate system, these rigid translations of the hBN layer both change the moiré structure and also shift it by an amount $\boldsymbol{d}_M$,

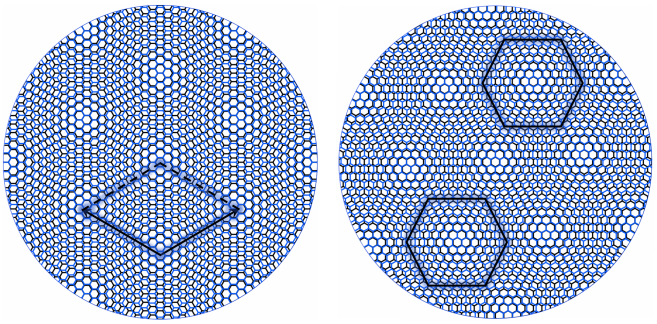
$$\boldsymbol{d}_M = (1 - \alpha \mathcal{R}_{\theta_{\text{GBN}}})^{-1}\, \boldsymbol{d}, \tag{35}$$

where α is the ratio of the lattice constants of our layers $\alpha = a_{BN}/a_G$ and $\mathcal{R}_{\theta_{GBN}}$ is a rotation counterclockwise by angle θ_{GBN}. The reason why this shift depends on both α and θ_{GBN} is because the moiré pattern G1/hBN is induced both by the mismatch between the lattice constants and by the twist angle between the layers.



(a) Layers and twist angles (b) moiré pattern when $\boldsymbol{d} = 0$ (c) moiré pattern when $\boldsymbol{d} \neq 0$

FIG. 10: Schematic illustration of the system: (a) side-view, (b) top-view in the case of AAA stacking and (c) top-view when the hBN layer is displaced by a vector $\boldsymbol{d}$. Diagrams (b) and (c) are representative for a commensurate tBG/hBN system and were taken from Shi *et al.* (2021) PhysRevB: 075122 [65].



(a) moiré pattern for a commensurate angle (b) moiré pattern for an incommensurate angle

FIG. 11: Examples of moiré patterns formed by two layers of similar lattice parameters, twisted at an angle.

By the end of this section we will show how this change in the moiré structure gives rise to different Chern numbers and therefore has an effect of the topological and also electronic properties of our system [65].

However, if we go slightly away from commensurability we lose translational symmetry, which means that superimposing our moiré patterns G1/G2 and G1/hBN gives rise to a supermoiré (a moiré of moiré) instead [67].

Considering very small variation angles from the commensurate points ensures that the supermoiré period is large enough for our structure to continue being characterised in terms of a local commensurate tBG/hBN, but with a shift parameter $\boldsymbol{d}$ slowly varying in space [65].

To better see this, we can look back at the zoomed region for the commensurate case in Figure 10(b). Clear zones of AAA stacking appear periodically at the centres of the moiré unit cells, despite the atoms of Carbon in G2 and Boron in hBN having different relative displacements from the Carbon atoms in G1, within the cell.

Either increasing or decreasing these commensurate twist angles by small amounts gives rise to a supermoiré. In this case, we can imagine a similar picture with $\boldsymbol{d} = 0$ at the origin, but now in each of the 6 moiré hexagons surrounding this middle one (like the two highlighted in Figure 11(b)), there is a small rigid displacement of the G2 and hBN layers from a local AAA stacking.

Because of the very small variation angles from the commensurate points, the supermoiré has such a large spatial periodicity that we are unable to see other regions of exact AAA stacking, as they lie outside our sample. This better explains why we claimed that our structure can still be roughly characterised in terms of a local commensurate tBG/hBN system.

B. Commensurate pairs of angles for tBG/hBN

The commensurability of the tBG/hBN trilayer is captured by the fact that any lattice vector of one moiré pattern can be written as a linear combination of the reciprocal basis of the other moiré. Since commensurability in real space implies commensurability in reciprocal space and vice-versa, we can write this condition as:

$$\boldsymbol{g}_1^{GBN} = \tilde{p}\,\boldsymbol{g}_2^{GG} + \tilde{q}\,\boldsymbol{g}_3^{GG}, \tag{36}$$

$$\boldsymbol{K}_{BN} - \boldsymbol{K}_1 = \tilde{p}\,\boldsymbol{q}_3 + \tilde{q}\,\boldsymbol{q}_2', \tag{37}$$

where $\tilde{p}$ and $\tilde{q}$ are rational numbers and all the vectors are shown in Figure 12. To go from (36) to (37), we rotate clockwise by 90° and scale by $1/\sqrt{3}$, such that we can explicitly write the components of the K_+ point of the first graphene layer $K_1 = (4\pi/3\,a_G, 0)$, and of hBN $K_{BN} = 4\pi/3a_{BN}(\cos\theta_{GBN}, \sin\theta_{GBN})$.

By looking at the same figure we can write $K_1 = K$, $K_2 = Ke^{i\theta_{GG}}$ and $K_{BN} = Ke^{i\theta_{GBN}}/\alpha$, such that by

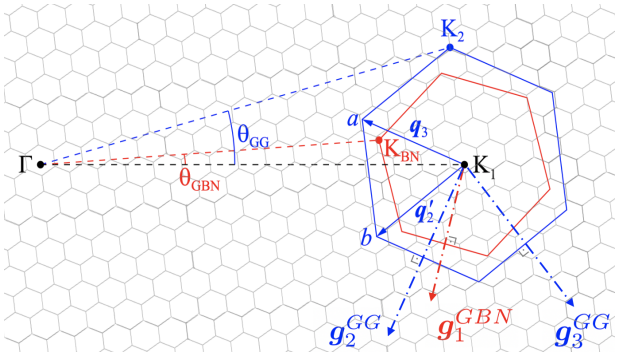


FIG. 12: Schematic illustration of the reciprocal space of tBG/hBN for generic angles θ_{GG} and θ_{GBN}. K_1, K_2, K_{BN} are BZ corners for the three layers, and the mini BZs of the G1/G2, G1/hBN and tBG/hBN moiré patterns are shown in blue, red and grey, respectively. While vectors $\boldsymbol{q}$ and $\boldsymbol{q}'$ point towards the corners of the tBG mBZ, reciprocal vectors $\boldsymbol{g}$ connect the centres of neighbouring mBZs. Adapted from multiple figures taken from Shi *et al.* (2021) PhysRevB: 075122 [65].

plugging these into Equation (37), we can exactly calculate pairs of commensuarte angles $(\theta^{\tilde{p}\tilde{q}}_{GG}, \theta^{\tilde{p}\tilde{q}}_{GBN})$. Despite being free to choose the values of $\tilde{p}$ and $\tilde{q}$, only three cases are discussed in this analysis:

$$\begin{cases} \tilde{p}{=}1, \tilde{q}{=}0 \rightarrow \theta^{60^\circ}_{GG} \approx 1.103^\circ, \theta^{60^\circ}_{GBN} \approx 0.551^\circ \\ \tilde{p}{=}0, \tilde{q}{=}1 \rightarrow \theta^{120^\circ}_{GG} \approx 1.116^\circ, \theta^{120^\circ}_{GBN} \approx -0.577^\circ \\ \tilde{p}{=}\tilde{q}=1/2 \rightarrow \theta^{90^\circ}_{GG} \approx 1.106^\circ, \theta^{90^\circ}_{GBN} \approx -0.009^\circ \end{cases}$$

which we are going to refer to as the 60°, 120° and 90° commensurate pairs, since these are the angles between segments $\overline{K_1K_2}$ and $\overline{K_1K_{BN}}$ which link the Dirac points of our layers in reciprocal space.

The reason why only these pairs were chosen is because they give twist angles θ_{GG} very close to the magic angle in tBG $\sim 1.05^\circ$ [28], where we expect to see strongly correlated behaviour [7]. These commensurate pairs were identified in Shi, J. *et al.* [65], but also in the previous works of Cea, T. *et al.* [61] and Lin, X. *et al.* [68].

C. Model Hamiltonian

Looking back to our initial setup, we anticipate that the tBG/hBN system will only depend on the free parameters θ_{GG}, θ_{GBN} and $\boldsymbol{d}$. Since the low-energy electrons lie in the graphene bilayer but feel the effect of the G1/hBN moiré pattern, we can model our system locally as

$$H(\boldsymbol{d}) = H_{tBG} + V_{BN}(\boldsymbol{d}), \tag{38}$$

by separating these two contributions. Note that the Hamiltonian H_{tBG} is a function of the twist angle θ_{GG}, and the potential V_{BN} depends both on the angle θ_{GBN} which gives rise to the moiré pattern and on the rigid displacement $\boldsymbol{d}$ which shifts it.

1. *AA Stacked Twisted Bilayer Graphene*

For twisted bilayer graphene we are going to consider the commonly adopted non-interacting model Hamiltonian first proposed by MacDonald, A. and Bistritzer, R. in 2011 [28]. This was already introduced in subsection II F, but we are going to revise it in more detail and rewrite the tunnelling contribution for the case of AA stacking this time.

Since we can obtain the Hamiltonian for the K_- valley valley by applying time reversal symmetry, it is enough to only focus on the K_+ valley:

$$H_{\text{tBG}} = \sum_{\boldsymbol{k}} \left(\psi^\dagger_{1k} \left(-\frac{U}{2} + \hbar v \boldsymbol{\sigma}_1 \cdot \boldsymbol{k} \right) \psi_{1\boldsymbol{k}} + \psi^\dagger_{2\boldsymbol{k}} \left(\frac{U}{2} + \hbar v \boldsymbol{\sigma}_2 \cdot \boldsymbol{k} \right) \psi_{2\boldsymbol{k}} \right) + \left(\sum_k \sum_{j=1}^{3} \psi^\dagger_{1\boldsymbol{k}} T_j \psi_{2(\boldsymbol{k}+\boldsymbol{q}_j)} + \text{ H.c.} \right). \tag{39}$$

The Hamiltonian is written in terms of field operators ψ and $\psi^\dagger$ acting on the Fock spaces of each of the graphene layers, which allow us to identify the first two terms as the Dirac Hamiltonians corresponding to G1 and G2, and the last one as a coupling term between the two layers of graphene.

The two Pauli vectors have the form $\boldsymbol{\sigma}_1 = (\sigma_x, \sigma_y)$ and $\boldsymbol{\sigma}_2 = (\cos\theta_{GG}\sigma_x - \sin\theta_{GG}\sigma_y, \sin\theta_{GG}\sigma_x + \cos\theta_{GG}\sigma_y)$, where the trigonometric terms in θ_{GG} account for the fact that the two layers are rotated by an angle with respect to each other. G1 and G2 also have an added potential difference U between them which can be changed externally, and v denotes the Fermi velocity at which the relativistic Dirac fermions move through graphene.

The last term sums over three processes in which an electron with momentum $\boldsymbol{k} + \boldsymbol{q}_j$, $j = 1, 2$ or 3, from the second graphene layer G2 hops to a state of momentum $\boldsymbol{k}$ in the first layer G1. While some of the vectors $\boldsymbol{q}_j$ are shown in Figure 12, the others can easily be inferred from there by noting that $\boldsymbol{q}'_j = -\boldsymbol{q}_j$. Each of these three processes happen with an amplitude given in the interlayer tunnelling matrices T_j:

$$T_j = \begin{pmatrix} w_{AA} & e^{-i\frac{2\pi}{3}(j-1)} w_{AB} \\ e^{i\frac{2\pi}{3}(j-1)} w_{AB} & w_{AA}, \end{pmatrix}, \tag{40}$$

where the matrix form arises from the fact that the

graphene layers each have two sublattices, A and B.

We can infer from the tunnelling matrix that an electron with momentum $\boldsymbol{k}+\boldsymbol{q}_j$ hops from sublattice A in G2 to sublattice A in G1 with amplitude w_{AA}, the same as an electron of the same momentum hopping from B in G2 to B in G1. However, depending on the vector $\boldsymbol{q}_j$, the amplitude w_{AB} of electrons hopping from sublattice A to B and vice-versa acquires a phase $e^{\pm i(j-1)2\pi/3}$.

As mentioned earlier in Subsection II E, graphite is naturally stacked in an AB configuration, as it is the most energetically stable structure [22]. Since in our analysis we consider an AA stacking instead, it is crucial to mention that our tBG system is subject to a lattice relaxation mechanism, which can be of two types.

The first is an in-plane atomic relaxation which wants to minimise the total energy of the crystal by decreasing the high stacking energy regions AA and increasing the AB/BA regions which have a lower stacking energy. The second is an out-of-plane atomic relaxation which has the effect of increasing the vertical separation between the AA regions, since the layer separation is shortest for local AB/BA stacking regions but longest for AA stacking.

An exact continuum model for low-energy electronic states in tBG proposed by Carr, S. et at. [69] allows us to approximately account for the effect of lattice relaxation by considering the ratio $w_{AA}/w_{AB} = 0.8$ between the two tunneling amplitudes [65].

2. G1/hBN Moiré Pattern

If we overlook for a second the lattice mismatch between graphene and hBN, we can easily picture that placing graphene on top of hBN produces an inhomogeneous electrostatic potential that breaks the sublattice/inversion symmetry. As mentioned before in Subsection II C, this results in a gap opening in the band structure [70].

Analogous to the Higgs mechanism in particle physics, breaking inversion symmetry in graphene gives rise to electrons effectively acquiring a mass m_0 [13]. Since the Hamiltonian for massive Dirac fermions in 2D has the form $H = v\boldsymbol{\sigma}\cdot\boldsymbol{p} + m_0\sigma_z$, where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y)$ as before, we expect a similar mass term in $V_{BN}(\boldsymbol{d})$ as well.

We assume that the effect of the G1/hBN moiré pattern is only felt by the first layer of graphene, such that V_{BN} is zero on G2 and non-zero on G1:

$$V_{BN}(\boldsymbol{d}) = \sum_{\boldsymbol{k}} \left(\psi^\dagger_{1\boldsymbol{k}}\,(m_0\sigma^z)\,\psi_{1\boldsymbol{k}} + \sum_{j=1}^{6}\psi^\dagger_{1\boldsymbol{k}} V_j(\boldsymbol{d})\psi_{1(\boldsymbol{k}+\boldsymbol{g}_j)}\right). \tag{41}$$

As discussed before, the first is a spatially averaged term responsible for breaking inversion symmetry and opening a gap in the spectrum, while the second accounts for the G1/hBN moiré pattern. Comparing the last terms in Equations (39) and (41), we can interpret the latter as an effective coupling between different sites in the reciprocal space of the G1 layer which differ by a momentum $\boldsymbol{g}_j$. These are transfer momenta from the first shell of moiré reciprocal lattices, i.e. from K_1 to the centres of the 6 neighbouring G1/hBN red mBZ in Figure 12.

The exact dependence on the rigid displacement $\boldsymbol{d}$ of the hBN layer is captured in the hopping matrix V_j by considering the Fourier coefficients

$$V_j(\boldsymbol{d}) = V_j(0)\, e^{i\boldsymbol{g}_j\cdot\boldsymbol{d}_M}, \tag{42}$$

where $\boldsymbol{d}_M$ is the amount by which the moiré pattern shifts and depends on $\boldsymbol{d}$ via Equation (35). $V_j(0)$ are constant matrices whose exact form depend on the rotational symmetries of the system [65].

All the constant quantities mentioned so far were calculated in multiple papers by Jung, J. *et al.* [71][72][73], who obtained different *ab initio* results under slightly different assumptions. The exact values chosen for this section can be found in Appendix C from Shi *et al.* (2021) PhysRevB: 075122 [65].

D. Quantum Anomalous Hall Effect at commensurate twist-angle pairs

Going back to our discussion from subsection II B, we can see that for any given Hamiltonian we can compute the Chern numbers using Equations (8) and (9).

Fukui, T. *et al.* proposed in 2005 [74] a manifestly gauge-invariant description of Chern numbers in an infinite system on the basis of a discretised Brillouin zone, and provided a highly efficient numerical method for computing Hall conductances without specifying gauge-fixing conditions.

By keeping the twist angles constant (for example considering the 60°, 120° or 90° commensurate pairs) and choosing a fixed potential difference U between the graphene layers, we can plot a map of the valence band Chern numbers as a function of the displacement $\boldsymbol{d}$. This is shown in Figure 13, together with the band structures at 3 different points in the map.

To better understand the colour map of Chern number in Figure 13, it is useful to look back at our initial initial setup in Figure 10(b). Because of the translational symmetry of the commensurate moiré lattice, we expect that shifting the hBN layer by a vector equal to the moiré wavelength has no effect on our system or its physics. As anticipated, the map of Chern numbers is also periodic in this case, since we can observe a recurrent structure created by the green and blue regions, which resembles a larger hexagonal structure.

Looking at Figure 10(c) now, we can see that a displacement $\boldsymbol{d}$ smaller than the one just considered gives a non-uniform shift from the AAA stacking for each point within the moiré unit cell. This is also captured in the colour map and explains why the regions we observe are not monochromatic.

It is however important to emphasise that in the absence of the second term in Equation (41), this map

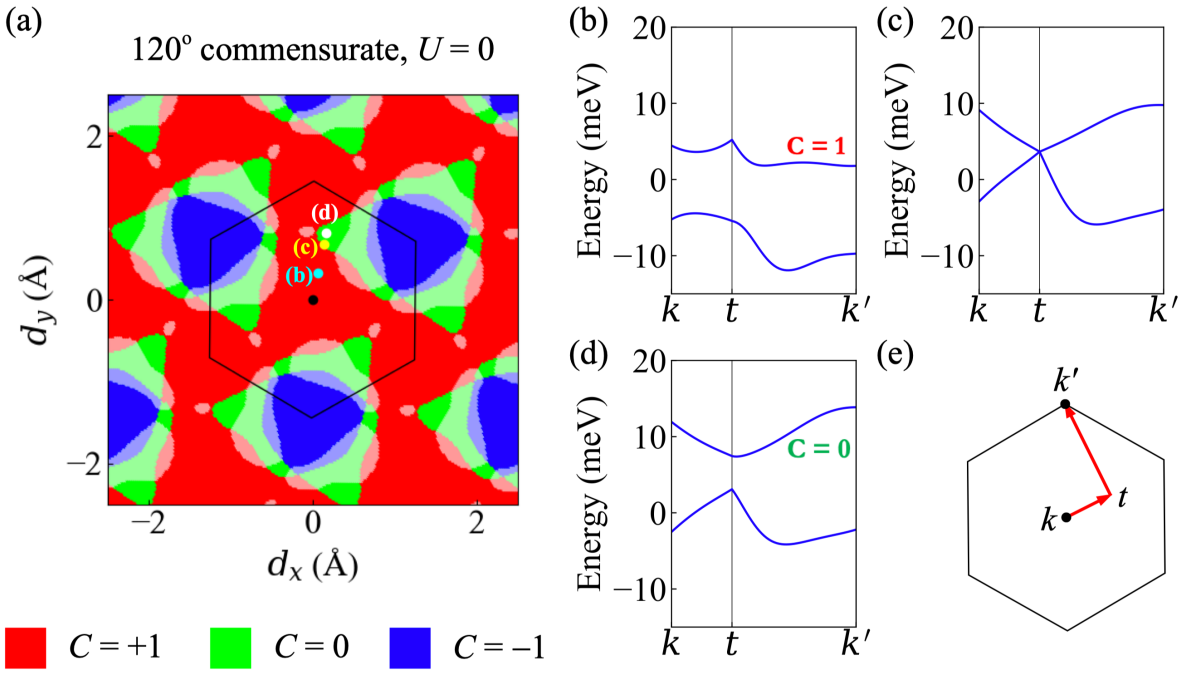


FIG. 13: Colour map of valence band Chern numbers as a function of the components of the hBN displacement $\boldsymbol{d}$, for a 120° commensurate tBG/hBN moiré system with zero interlayer potential difference U. (b), (c) and (d) are the band structures corresponding to the marked dots on the map, taken along the path (e) in the mBZ, where t labels the band touching point. The lighter shades are semimetal regions and the black hexagon is the Wigner-Seitz cell of the hBN layer in real space. Image was taken from Shi *et al.* (2021) PhysRevB: 075122 [65].

would only have one colour, as there would be no $\boldsymbol{d}$ dependence in the Hamiltonian. This interesting structure is only present because of the qualitative influence of the G1/hBN moiré pattern on the graphene bilayer.

Subfigures 13 b), c) and d) show the band structures of the system at the exact displacements emphasised by the 3 points in cyan, yellow and white in the colour map. It can clearly be seen that going from a Chern insulator state with $C = 1$ to a trivial insulator state characterised by $C = 0$, the band gap closes and reopens, such that we get a band inversion where the valence and the conduction band exchange Chern numbers [65]. The gap is only present in the spectrum at specific points in the (d_x, d_y) map which are on the interface between different colours, since in the absence of a gap those states cannot be topologically classified by one Chern number.

Another crucial thing to note in Figure 13 are the lighter shades on the colour map, which in fact correspond to semimetal regions with a gap closing that is indirect in momentum space. Since in a semimetal the top of the conduction band and the bottom of the valence band overlap, this results in a gap closing (either direct or indirect), and therefore cannot support a quantised Hall conductance.

In practice, this means that devices apparently identical in terms of the (commensurate) twist angles and the interlayer potential may or may not support a quantised QAHE, depending on the exact displacement of the hBN layer with respect to an AAA stacking configuration.

E. Quantum Anomalous Hall Effect at nearly commensurate twist-angle pairs

Although a beautiful result, this cannot be tested experimentally since the current precision for controlling θ_{GG} and θ_{GBN} is of order $\sim 0.1°$ and $\sim 1°$, respectively.

Comparing these possible angular variations with our estimations for the commensurate pairs of angles given in subsection IV B, we can see that such a commensurate moiré structure cannot be faithfully engineered in a device, since even small angle deviations take us away from commensurability.

We now considered the same system but with twist angles slightly away from the commensurate points $(\theta^{\tilde{p}\tilde{q}}_{GG}, \theta^{\tilde{p}\tilde{q}}_{GBN})$ discussed in IV B:

$$\theta_{GG} = \theta^{\tilde{p}\tilde{q}}_{GG} + \delta\theta_{GG} \text{ and } \theta_{GBN} = \theta^{\tilde{p}\tilde{q}}_{GBN} + \delta\theta_{GBN}, \quad (43)$$

where we take $\delta\theta_{GG}, \delta\theta_{GBN} \sim 0.01°$, which ensures that the supermoiré period is large enough for us to still use the same local commensurate approximation.

Unlike the commensurate case when a rigid displacement of the hBN layer changes and shifts the moiré pattern - Figures 10(b) and 10(c) - shifting the hBN layer by a certain amount does not qualitatively change the

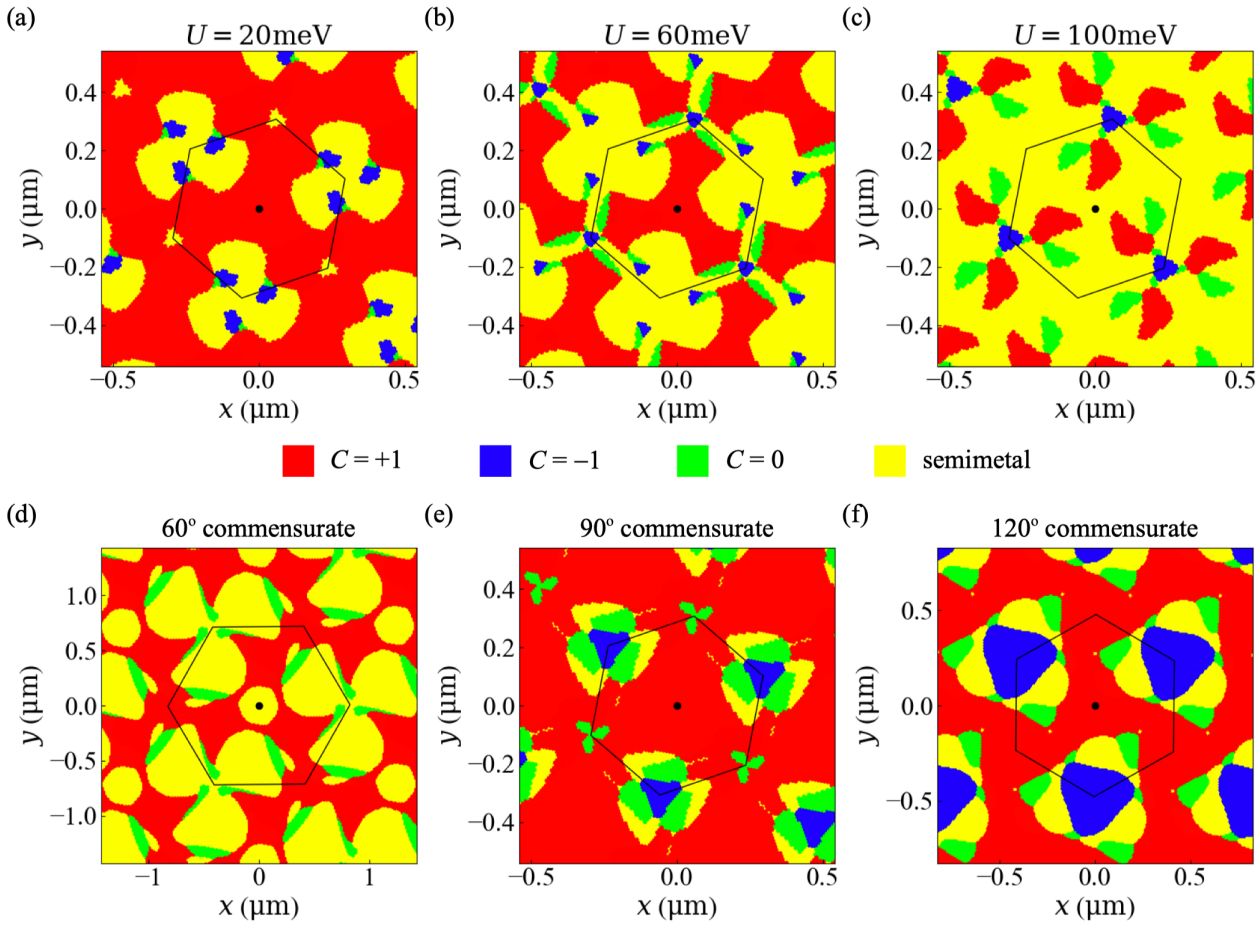


FIG. 14: Colour maps of the valence Chern numbers in supermoiré structures as a function of spatial coordinates. In (a), (b), (c) we considered a system close to a 90° commensurate twist point, $\delta\theta_{GG} = \delta\theta_{GBN} = 0.01°$, and varied the interlayer potential difference U. In (d), (e), (f) we considered a supermoiré close to different commensurate points, $\delta\theta_{GG} = \delta\theta_{GBN} = 0.01°$, and kept the interlayer potential difference at $U = -25$meV. The black hexagon is the Wigner-Seitz cell of the supermoiré pattern. Image was taken from Shi *et al.* (2021) PhysRevB: 075122 [65].

supermoiré. A rigid displacement $\boldsymbol{d}$ of the hBN layer causes a magnified shift of the G1/hBN moiré pattern, which in turn produces a further magnified shift of the supermoiré pattern, but this can be effectively cancelled by reselecting the origin [65].

Thus, we no longer expect for the Chern numbers to depend on the displacement $\boldsymbol{d}$ of the hBN layer. However, as explained before in subsection IV A, a supermoiré can be seen as a local commensurate with a shift parameter slowly varying in space. This means that we can map our Hamiltonian from $H(\boldsymbol{d})$ into $H(\boldsymbol{d}(\boldsymbol{r})) \equiv H(\boldsymbol{r})$, which allows us now to calculate Chern numbers as a function of space coordinates.

Hence we expect our supermoiré structure in real space to be characterised by complex, spatially inhomogenous states of alternating Chern insulator ($C \neq 0$), trivial insulator ($C = 0$), and semimetal phases (cannot assign a Chern number to the valence band). In Figure 14 we can see 6 distinct colour maps of valence band Chern numbers for supermoiré structures of different twist angles and interlayer potentials.

In order to relate these colour maps to topological electronic properties, such as the QAHE, we need to interpret them using a percolation-like picture. Trugman, S. in 1983 [75] and Chalker, J. together with Coddington, P. in 1988 [76] were the first ones to describe the quantum Hall effect as a percolation problem.

We can see that in Figure 14, subfigures (a), (d), (e) and (f) each have a globally connected red phase, which we are therefore going to call percolating. Since the red phase corresponds to a nonzero Chern number $C = 1$, this indicates that an overall measurable QAHE can be found in each of these cases, as topologically nontrivial phases percolate across the device [65].

However, since in subfigures (b) and (c) the yellow semimetal phase is percolating, we can conclude that a QAHE cannot be supported at the given interlayer potentials $U = 60$meV and $U = 100$meV in a nearly commensurate 90° supermoiré structure.

Intuitively, this can be easily understood by thinking

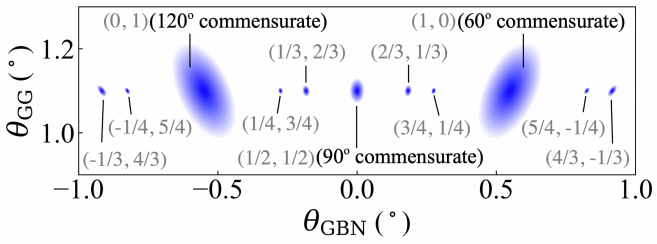


FIG. 15: Twist angle windows for QAHE, where the shaded blue regions show where the Hall conductance is expected to be quantised. These regions are labeled by the pair of rational numbers $(\tilde{p}, \tilde{q})$, shown in grey. Image was taken from Shi *et al.* (2021) PhysRevB: 075122 [65].

that having for example a metal phase percolating means that electrons can always find a path within the sample to scatter from the source to the drain attached to the device, such that a conductance can be measured.

This can be intuitively understood by imagining that having a specific phase percolating across the device, for example a metallic phase, means that electrons can always find a path within the sample to scatter. Therefore, attaching a source and a drain on the boundary of the device will give a measurable conductance, since a net electron flow between them is ensured by the metallic percolating phase.

By the same logic, a percolating phase of nonzero Chern number, $C = 1$ or $C = -1$, allows us to understand the QAHE using a similar transport picture. Therefore, we argue that the presence or absence of the QAHE in a system can be verified by looking at the percolating phases in maps of valence band Chern numbers. Note that the range in the x and y axis is chosen to be of the same order of magnitude as the dimensions of sample devices produced experimentally [65][77].

We can now extend this analysis for all pairs of twist angles within our experimental precision $\delta\theta_{GG} \sim \pm 0.1°$ and $\delta\theta_{GBN} \sim \pm 1°$, more specifically $\theta_{GG} \in (1.0°, 1.2°)$ and $\theta_{GBN} \in (-1.0°, 1.0°)$. Looking at the percolating states of each of these cases allows us to estimate the exact twist angle windows within which QAHE can occur in tBG/hBN supermoiré. This is explicitly shown in Figure 15 by the blue shaded regions.

For smaller rational numbers $\tilde{p}$ and $\tilde{q}$, the tBG bands are expected to split into a large number of subbands, i.e. for $\tilde{p}, \tilde{q} \sim 1/n$, each moiré band of isolated tBG splits into n^2 bands. This is unlikely to lead to an observable QAHE [65], so the $(\tilde{p}, \tilde{q})$ pairs chosen in Figure 15 are enough to characterise the twist angle windows.

Direct verification of this theoretical proposal could be achieved by performing and transport measurements in the same sample [65].

V. CONCLUSIONS AND OUTLOOK

As discussed in Section III, we can conclude that looking at the topological properties of a system allows us better characterise it and to even differentiate between configurations which were impossible to distinguish experimentally before. Despite the fact that the two stacking configurations AB-AB and AB-BA of tDBG have similar band structures and consequently, almost identical transport properties, they are topologically distinct.

In the absence of a perpendicular electric field, AB-AB is topologically trivial, while AB-BA is a valley Hall insulator, when the Fermi energy lies in a gap. Experimental measurements have found unique topological quantum Hall states in AB-BA, which further confirms the topological distinction between the two stacking configurations. The Landau level spectrum can highlight these topological differences, as a result of degeneracy lifting in the AB-BA configuration.

These findings indicate that the stacking chirality is a new degree of freedom available in tDBG, which may extend to other multilayer moiré systems. This could provide a new ingredient of control when designing new, device-focused vdW heterostructures.

In the trilayer tBG/hBN superlattice system discussed in Section IV, we showed that when the hBN layer is nearly aligned with the adjacent graphene layer, the new moiré pattern produces an effective external potential on the low-energy electrons localised in tBG. This potential contains both a position independent and dependent term, the latter often being ignored in the theoretical descriptions of the system.

When both terms are considered, valence band Chern numbers of commensurate trilayers change as the hBN layer is shifted by a rigid displacement $\boldsymbol{d}$. However, when the position-dependent term is omitted, the Chern number colour maps are monochromatic.

Building on this result, the tBG/hBN trilayer is also studied at twist angles close to commensurability, when a long-period supermoiré pattern is formed. Looking at the spatial Chern number maps for different twist angles, we argue that an overall QAHE is only present when a topologically nontrivial insulating phase percolates.

This analysis is extended to variations in the twist angles within the current experimental precision, which allows us to find twist angle windows where the QAHE is expected to be present. This approach can explain the experimental observation of both quantised and non-quantised AHE in tBG/hBN structures, as well as states with no AHE at all.

To conclude, topology is a powerful and fundamental concept for characterising phases of matter in 2D materials, which was exemplified in detail for the cases of tDBG and tBG/hBN systems. The ability to host topologically interesting phases, such as new degrees of freedom available in multilayer moiré systems and correlated insulating states with various Chern numbers, makes them a promising platform for further research.

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